# Design and Evaluation of an Augmented Reality Storybook for Primary Learners' Plant Growth Understanding & Stewardship

Project ID: R26-IM-001

# Problem Statement and Stakeholders

**OBSERVED ISSUE**

- Traditional text-based teaching fails to convey dynamic plant growth to primary learners.
- Current AR solutions often causes cognitive overload and lack engaging storytelling, and ignore real-world environment context, leaving kids disconnected from hands-on ecological literacy.

**WHY IT MATTERS**

- Bridging real-world conditions with virtual learning.
- Creating a safe, child-friendly alternative to complex VR setups.
- Making plant science interactive, helping children build long-term environmental awareness and responsibility.

**Primary Users:**

Primary learners (Age 6-8) who need:

- Interactive plant learning experiences.
- Better understanding through AR-based feedback.

**Secondary Users:**

- Teachers, Parents
- Educational institutions
- AR educational researchers
- Environmental learning developers

# Research Gap & Objectives

Existing AR learning systems lack adaptive and child-friendly interaction methods for physical book-based learning.

Current AR plant simulations lack emotional visual feedback to represent different plant health conditions.

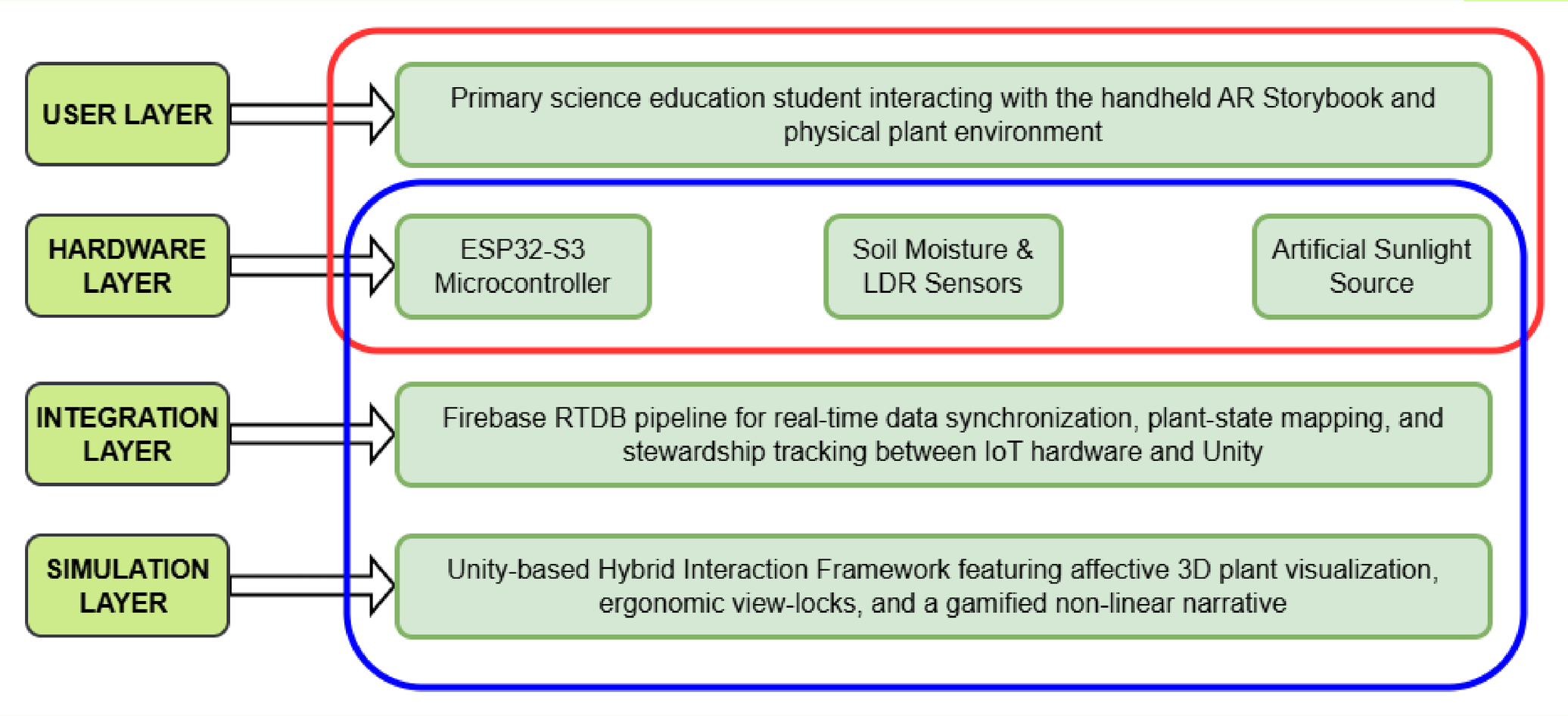
story narratives in current educational AR remain fixed and fail to adapt dynamically to a learner's actions or plant states.

Most AR tools operate in isolation, missing real-world environmental data to dynamically influence virtual plant growth behavior

**Main Objective:**

To develop an AR storybook framework that integrates adaptive interaction, affective plant visualization, gamified storytelling, and context-aware environmental simulation to create an engaging and realistic plant growth learning experience for primary learners.

# System Architecture & Integration



# Research Questions

### HYBRID INTERACTION FRAMEWORK

How can a child-centered hybrid interaction framework mitigate touch inaccuracy, AR instability, the "Gorilla Arm Effect," and cognitive load to enhance the usability and learning effectiveness of handheld AR storybooks for Grade 1 learners?

### AFFECTIVE VISUALIZATION MODEL

How can an AR storybook clearly and emotionally communicate plant health, stress, and recovery states to young children in a way that improves understanding and encourages empathy toward plants?

### GAMIFIED NARRATIVE ENGINE

How to Design a Gamified Narrative Engine That Builds Children's Environmental Responsibility Through AR Storytelling?

### CONTEXT-AWARE SIMULATION ENGINE

How to design a Context-Aware Simulation Engine respond to real-world environmental conditions and user interactions in real time ?

# Solution Overview

| Hybrid Interaction Framework | Affective Visualization Model | Gamified Narrative Engine | Context-Aware Simulation Engine |
|---|---|---|---|
| Dynamic UI Resizing to reduce touch difficulty. | State-based plant visualization model representing Healthy, Stressed, and Recovery states. | Stewardship-based narrative progression where plant care affects story development. | ESP32-based IoT environmental monitoring using DHT22, LDR, SI1145, and Soil Moisture sensors. |
| Stable AR anchoring to reduce cognitive workload. | Uses colour changes, posture changes, animation, and visual effects to communicate plant conditions. | Adaptive story branching based on learner choices and plant conditions. | Firebase real-time synchronization between hardware and Unity AR application. |
| AR-to-2D transition for Gorilla Arm Effect. | Provides emotional feedback to improve plant care understanding. | Recovery mechanism allows children to correct negative actions through care. | Dynamic plant growth behaviour based on environmental conditions and user interaction. |

# Commercialization & Impact

## VALUE PROPOSITION

A curriculum-aligned AR storybook combining:

- Physical storybook learning
- Interactive AR visualization
- Emotional plant state feedback
- Real-world environmental response

## TARGET USERS

- Grade 1 teachers and primary learners.
- Parents supporting guided home learning.
- International schools.
- Educational institutions seeking interactive science learning solutions.

## ADOPTION PLAN

**Phase 1: Pilot Implementation**

- Deploy prototype in selected primary classrooms and educational environments.

**Phase 2: Educational Integration**

- Integrate with schools using Cambridge Primary Science curriculum.

**Phase 3: Commercial Expansion**

- Expand into:
    i. Private schools
    ii. Home-learning AR kits
    iii. Additional AR science learning modules